

5CM510 – Network System development

Lecture 21: Quantum Networking

Lecture Outline

- Quantum Communications protocols and Networking Preliminaries
- Quantum Routing
- Quantum Network Simulators
- Summary

Quantum Communication and Networks

- Quantum computing operates on quantum bits (qubits) encoded in the quantum state of particles such as photons, electrons, and atoms.
- A quantum state has the particularity of being in a superposition, i.e., 0 and 1 at the same time.
- Various technologies can be used to represent a qubit such as encoding a quantum state in photon polarization or electron spin. The principles of quantum mechanics hold regardless of the underlying qubit technology.
- Scaling the computational power of quantum computers and expanding the potential of quantum computing to more applications requires the transfer of qubits between distant nodes without destroying their superposition states.
- Quantum communication is made possible through entanglement, a phenomenon where distant particles (e.g., photons) exhibit strongly correlated behaviors in a way that defies classical physics.
- For example, Alice begins by encoding her message into the quantum states of photons, which will be sent to Bob. This initial step introduces the challenge of maintaining the integrity of the qubits during transmission.

Quiz 1: Quantum Communication and Networks



What is a qubit, and how does it differ from a classical bit?

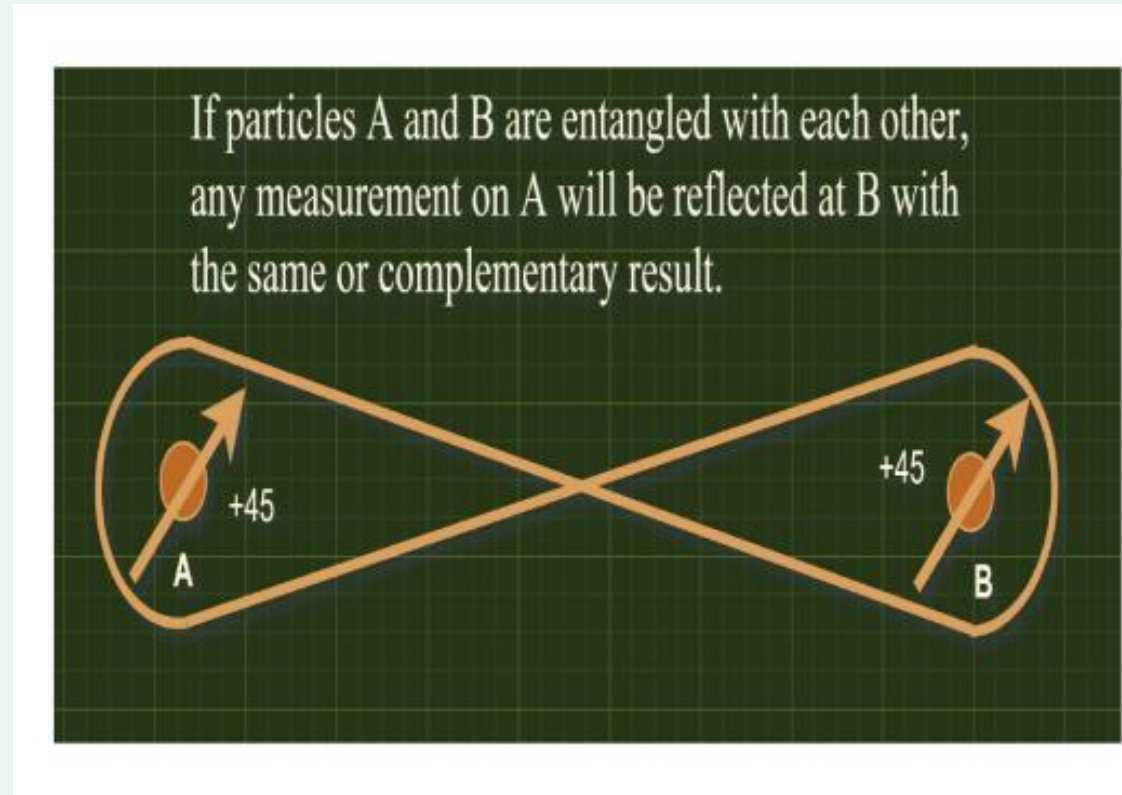


Define quantum entanglement and provide one application.

No-Cloning and Decoherence principle

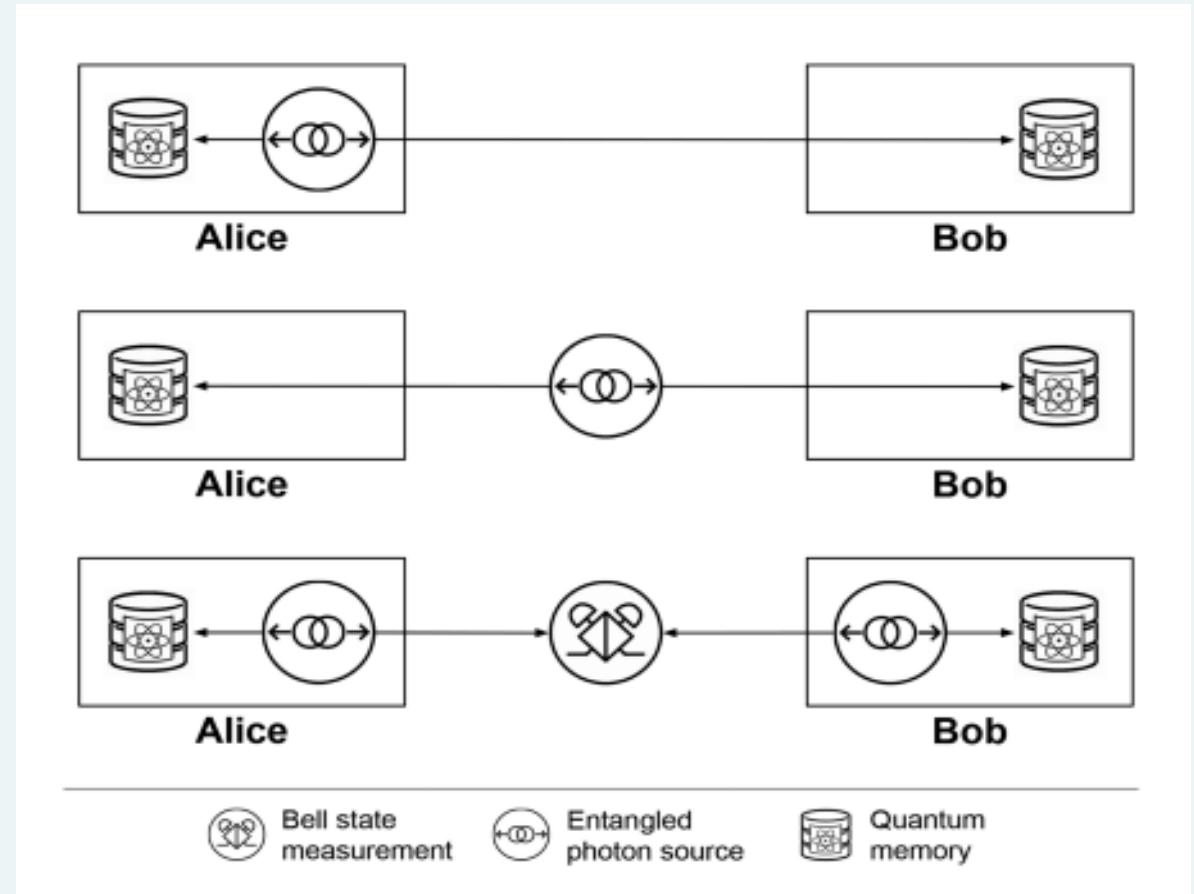
- The no-cloning principle prohibits the exact replication of quantum states, ensuring that quantum information remains fundamentally secure but also challenging to manipulate.
- Any measurement of a quantum system inevitably alters its state.
- Consequently, qubits cannot be measured and then replicated to repeat a quantum signal or retry a transmission. These principles do not hold in with classical signals, which is why classical networks can use amplifiers, feedforward error correction, and retransmission to reliably send units of data across the network.
- Longer wait times can increase the likelihood of decoherence (“Quantum weirdness”), degrading the fidelity of entangled states and increasing the error rates.
- If the wait time exceeds the coherence time of the memory, the stored qubits will become corrupted, rendering them useless. However, if the coherence time is sufficiently long, policies with longer wait times may achieve higher success rates by making better swapping (and possibly purification) decisions, thus improving E2E entanglement throughput.

Quantum Communication: Entanglement



Entanglement generation and heralding

- One way to overcome the limitations of nondeterministic sources is by heralding.
- Heralding refers to the process of mutually acknowledging the confirmed presence or absence of an entangled pair between the nodes attempting an entanglement creation.
- For example, one can implement a heralded entangled photon source by generating bipartite entangled states conditioned on the detection of additional auxiliary photons.
- If these auxiliary photons are found to meet the requisite conditional measurement patterns, they herald the successful creation of the entangled pair.
- Another example scheme would be to integrate heralded quantum memories at each end node, which can confirm the successful storage of a photon upon its arrival.

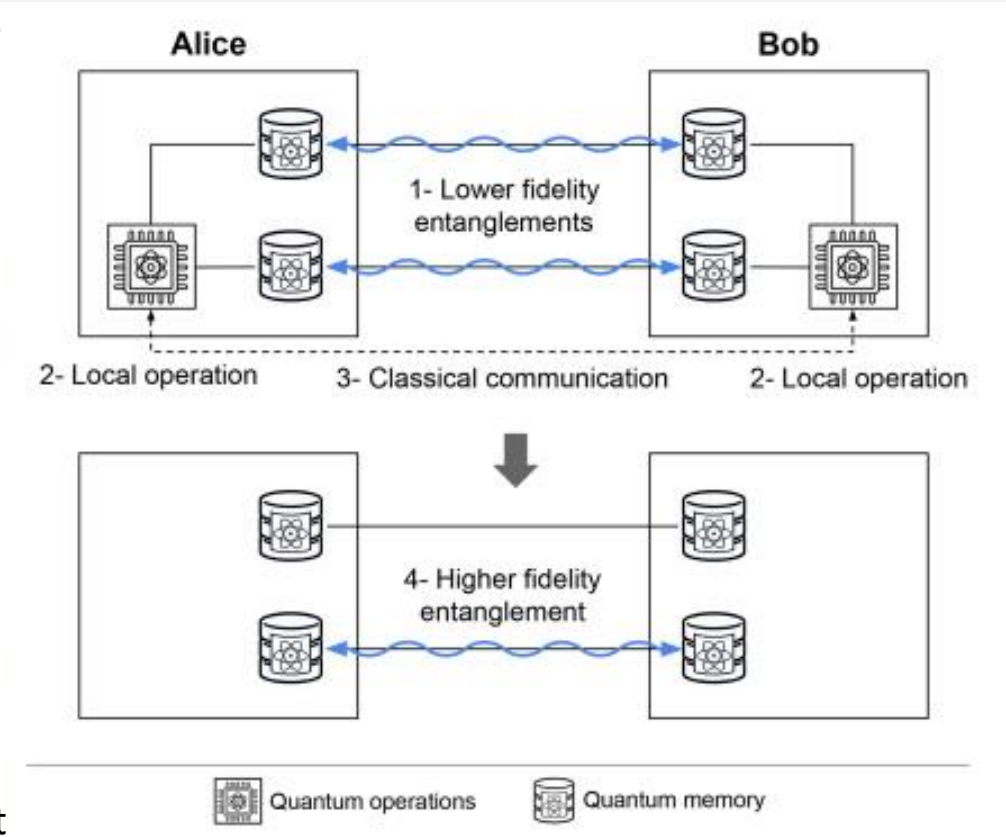


Quantum Memory

- Entanglement generation schemes frequently use photons (flying qubits) to encode quantum information and deliver the generated entangled states to end nodes over optical fiber, and atoms, or matter qubits, to store the entangled state at each node in a quantum memory.
- Photons are a promising medium for encoding and transmitting quantum information due to their degrees of freedom and their fast traversal speed over fiber and free-space optical links.
- Capturing and storing a photon in a quantum memory can be achieved with various technologies, either with solid-state platforms or free-space platforms. Some platforms support on-demand photon emission from the memory, and others may only emit at set time intervals.
- Memories can be characterized by key parameters such as their storage time representing the time interval beyond which the stored quantum state is irreversibly degraded and can no longer be used.
- This results from entanglement decoherence, where the entangled pair of particles degrades over time due to interactions with their surrounding environment. Other memory parameters can be storage efficiency, retrieval efficiency, and an acceptable wavelength range.

Entanglement Purification

- Decoherence, fibre loss, and noisy quantum operations prevent entangled states from achieving or maintaining optimal quality entanglement with respect to a desired state, thus compromising the utility of entanglement links in the network.
- The probability that generated entanglements are in a certain desired state is quantified by a value known as the fidelity. Entanglement purification (or distillation) provides an efficient means of generating high-fidelity entangled states from multiple copies of noisy entangled pairs.
- Unlike error correction, protocols assume that the target state is known. Purification is particularly critical in long-distance communication with quantum repeaters, where it poses a bottleneck due to substantial overhead that affects transmission rates and error resilience.
- Entanglement purification protocols use local operations and classic communications (LOCC) to reduce noise in an ensemble of entangled pairs, enhancing the entanglement of a subset while discarding the rest.
- Protocols work by using some of the entangled pairs to collect information about others without consuming them. This information enables the protocol to identify, filter, and sometimes correct errors within the remaining pairs, effectively “purifying” the ensemble.

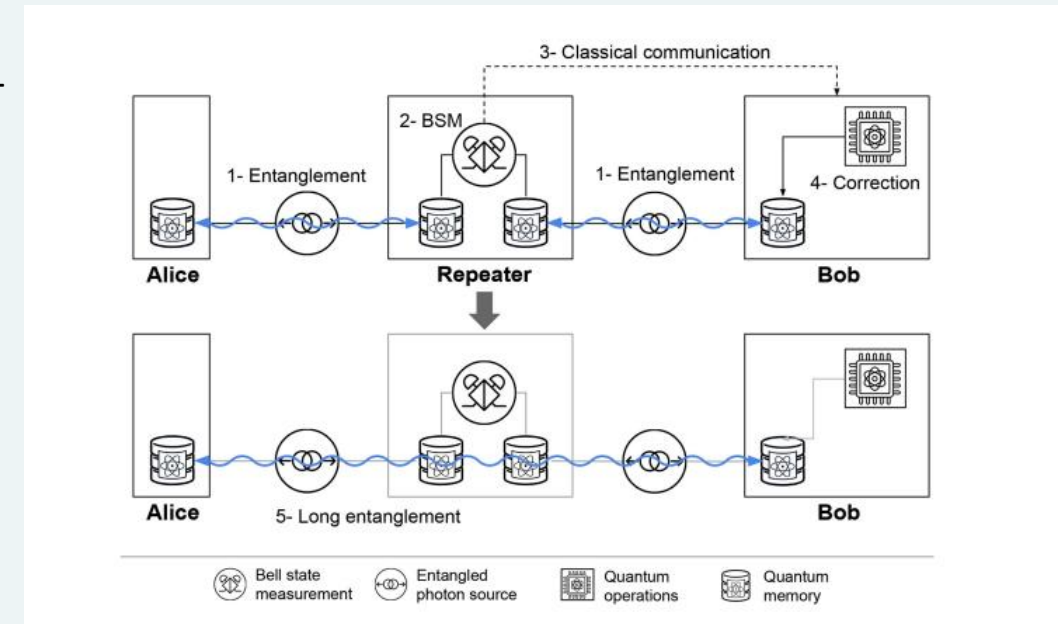


Quantum Error Correction(QEC)

- In addition to purification, QEC is essential to manage errors in quantum states during transmission.
- QEC techniques encode a logical qubit into a block of several physical qubits, which helps to preserve the integrity of the quantum state despite potential errors that may arise from decoherence or other quantum operations.
- By encoding and then decoding the information, QEC allows for the correction of errors without needing to directly observe the quantum state, thereby not violating the no-cloning principle.
- However, the effectiveness of QEC is restricted by the no-cloning theorem to less than a 50% threshold.
- This error threshold represents the maximum rate of errors that can be corrected using a QEC protocol. Beyond this threshold, the accumulation of errors outpaces the capability of the error correction process, failing to recover the original quantum information

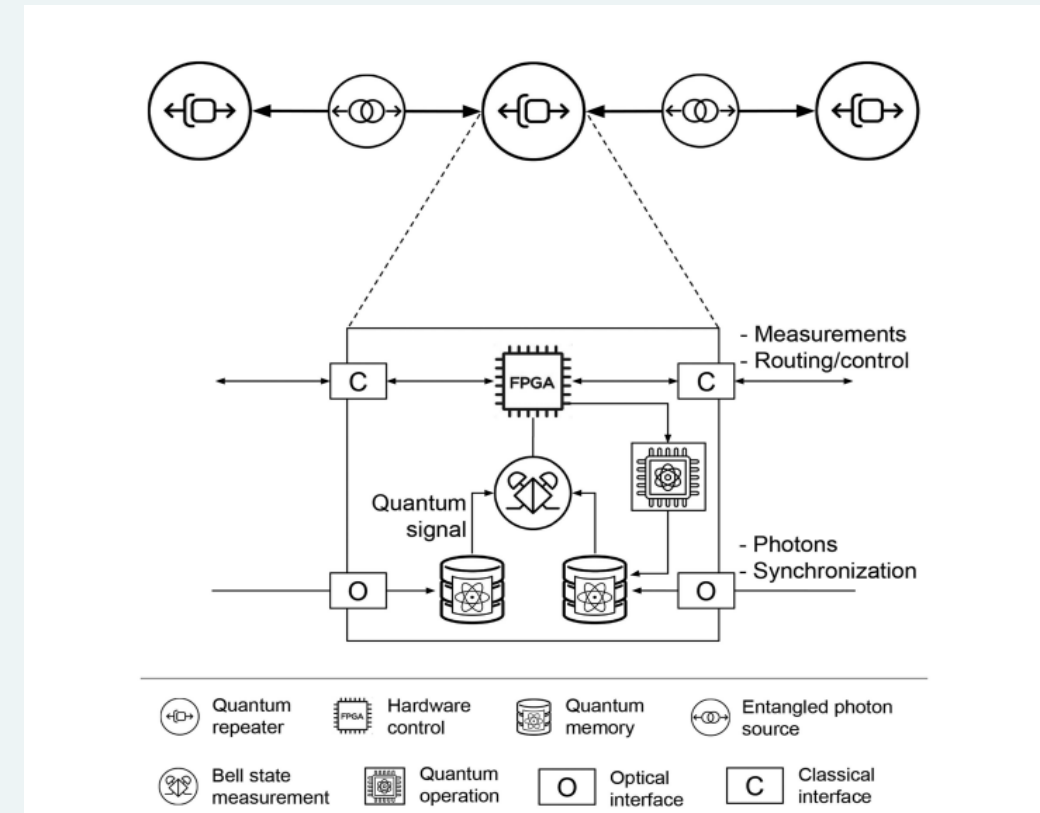
Entanglement Swapping

- Inherent lossy nature of optical channels: the probability of successfully distributing an entangled pair decreases exponentially as the physical distance of the channel increases.
- Methods for mitigating photon loss to extend the distance over which high-quality entanglements are distributed.
- using high-rate sources or multiplexing techniques (increase the number of attempts per unit time), high-efficiency photon detectors (more accurate measurements), low-loss optical fiber, or free-space channels can all help improve the hashing with entanglement gathering (HEG) success probability.
- The swapping process is one repeater is placed between Alice and Bob to split the E2E distance into two smaller distances. Two elementary entanglements are generated; Alice → repeater and the repeater → Bob. The repeater then measures its two entangled local qubits and sends the measurement outcome to Bob. Based on the received measurement, Bob applies a correction to its local qubit, creating an E2E entanglement between Alice and Bob. The success of the swapping operation is also probabilistic. To support longer distances between Alice and Bob, more repeaters can be placed between the two end nodes.



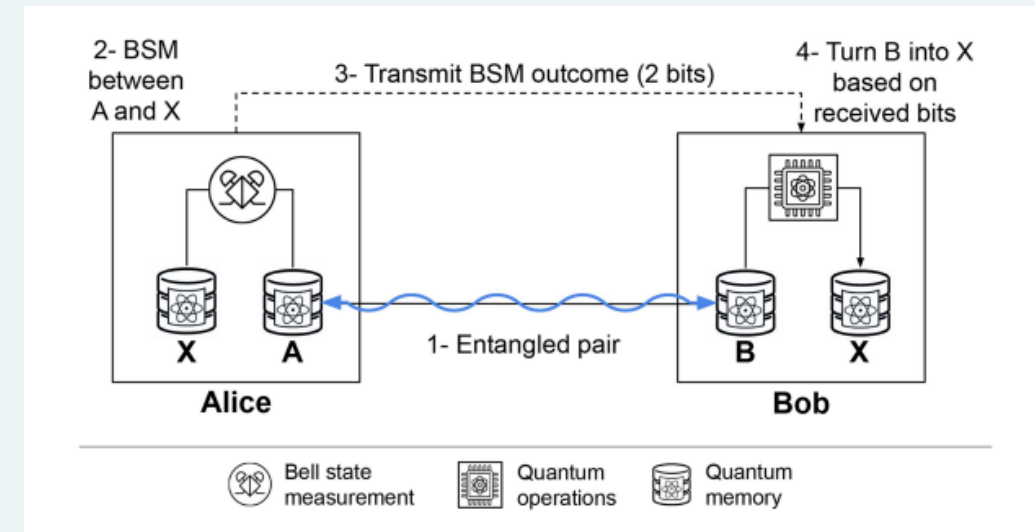
Quantum Repeaters

- The main function of a repeater is to implement the swapping process described earlier. Entanglement distribution over quantum repeaters is subject to two main types of errors:
- loss errors arising from the attenuation of fibres and
- operational errors resulting from inaccuracies in manipulating and measuring quantum states.
- By repeating the heralded process until the neighbouring repeaters confirm the successful creation of an entanglement, the HEG protocol serves as a mechanism for natively mitigating loss errors. During this period, the entangled qubits are maintained by the repeaters until a clear indication of success or failure is communicated.
- On the other hand, QEC offers a more efficient approach, though it demands more resources. At each repeater, QEC can be implemented to restore the original logical qubit.
- The quantum and classical signals flow unidirectionally, that is, the end nodes do not store any quantum states once the logical qubit is restored and transmitted.

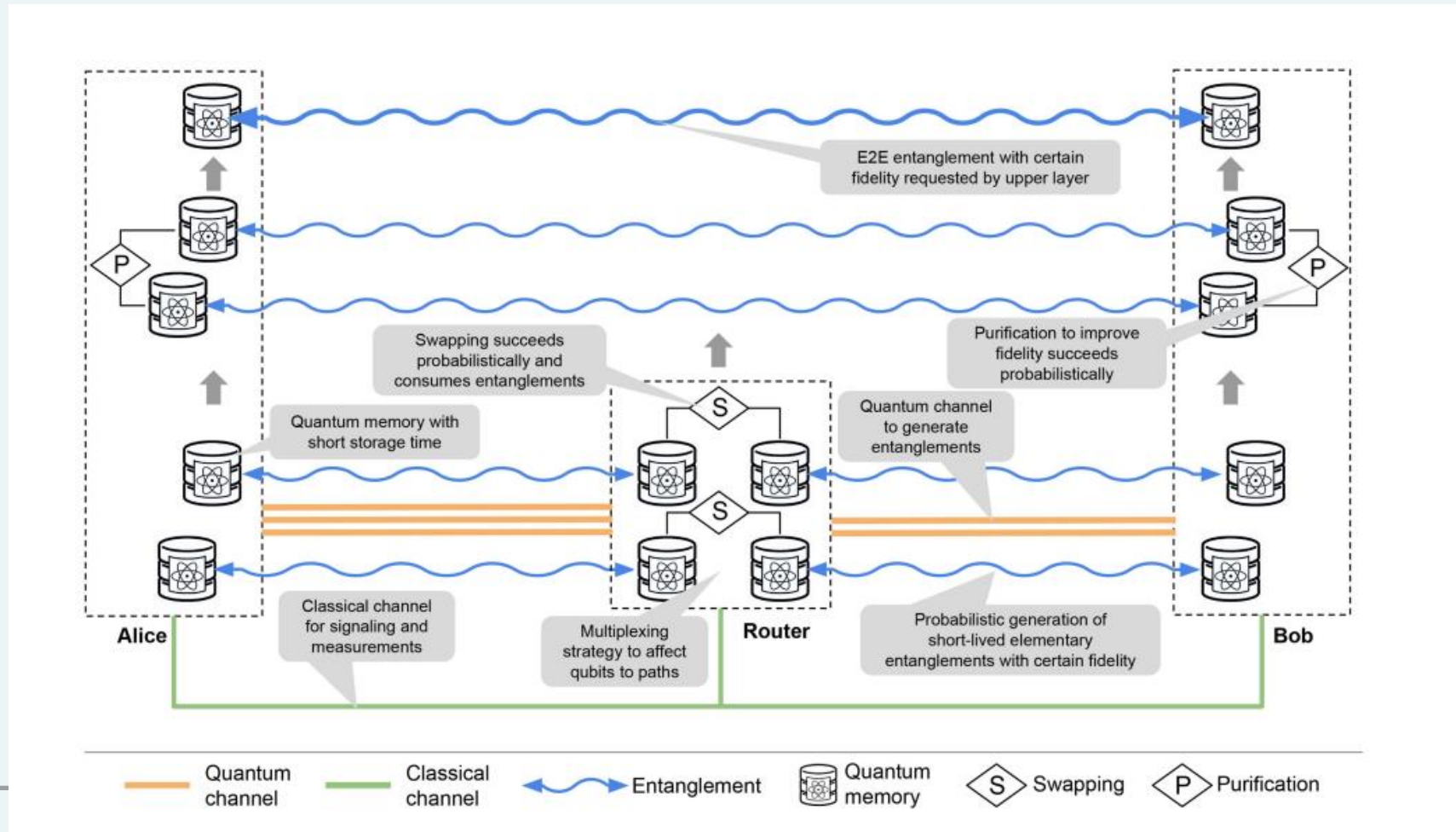


Quantum Teleportation

- Distributing entanglements between Alice and Bob does not constitute an exchange of quantum data by itself.
- Instead, it serves as a connection to transmit data qubits through the teleportation process. Since qubits cannot be copied, teleportation is used for the transmission of a quantum state without physically transferring the quantum particle that encodes the qubit.
- The teleportation process is depicted in the procedure for teleporting an arbitrary qubit X from Alice to Bob involves generating an E2E entanglement between Alice and Bob, where qubit A is located at Alice and qubit B at Bob. Alice then performs a BSM of X and A, and the two-bit result of this measurement is communicated to Bob through a classical channel.
- Upon receiving this information, Bob applies specific operations to qubit B based on the measurement outcome, resulting in the retrieval of qubit X.
- While the teleportation operation is described to illustrate an E2E exchange of quantum information, it is worth noting that there are applications, such as entanglement-based quantum key distribution, that consume E2E entanglements directly without teleportation.



Quantum Communication protocols for E2E Entanglement requests



Quiz 3: Quantum Networking Components



What role do quantum repeaters play in quantum networking?

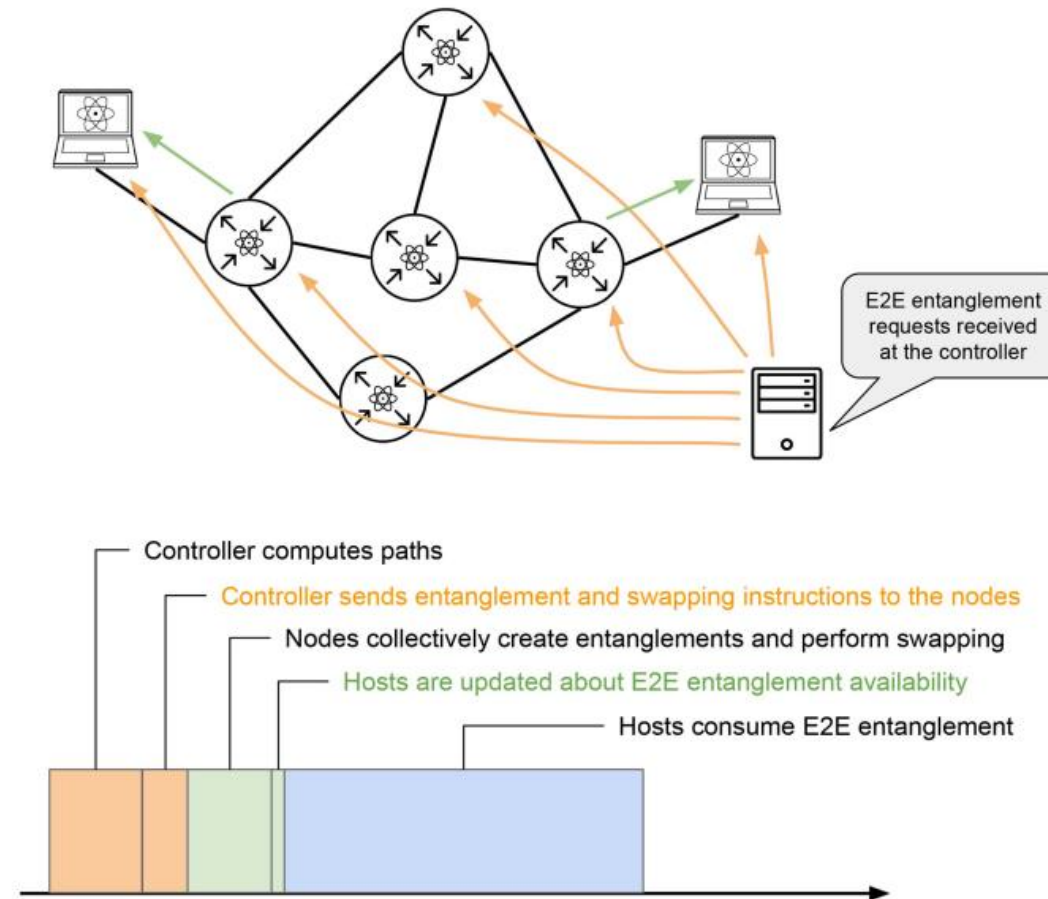


Describe entanglement swapping and its significance.

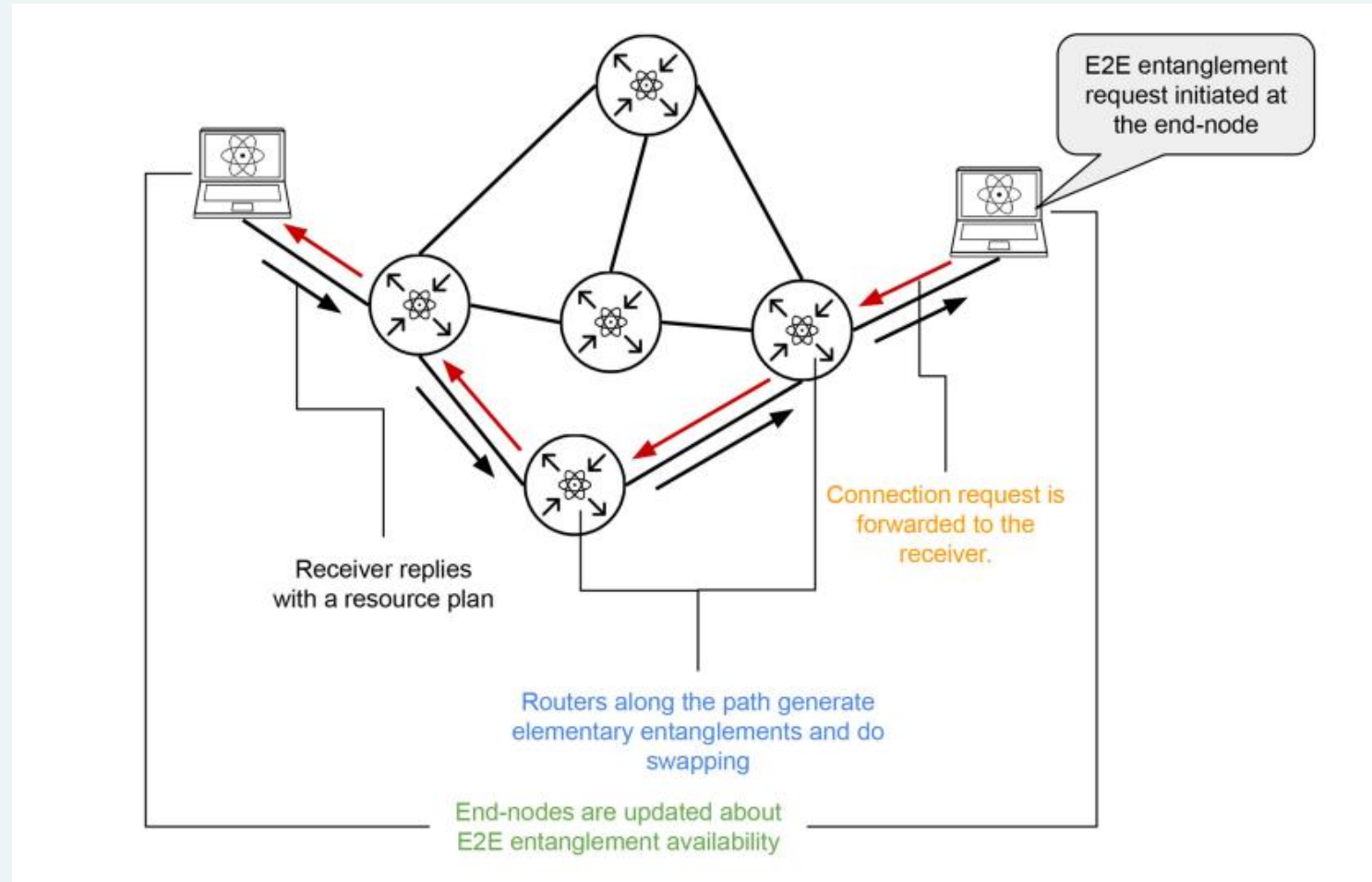
Quantum Routing - Proactive

- In proactive routing (sometimes designated as on demand entanglement generation) the path computation and route installation take place before elementary entanglements are created.
- A centralized proactive routing architecture is illustrated as E2E entanglement requests are received by the controller, which computes the paths and provides the nodes (routers and end nodes) with instructions for entanglement creation and swapping for each path.
- Subsequently, the nodes coordinate to create E2E entanglements, incorporating purification processes if supported. End nodes are notified of available E2E entanglements as part of the swapping process.
- Distributed proactive routing is also possible and uses a connection phase to install paths and negotiate resources between nodes. Path computation can be done at the requester or the receiver, or hop-by-hop along the path.

Centralized Proactive routing



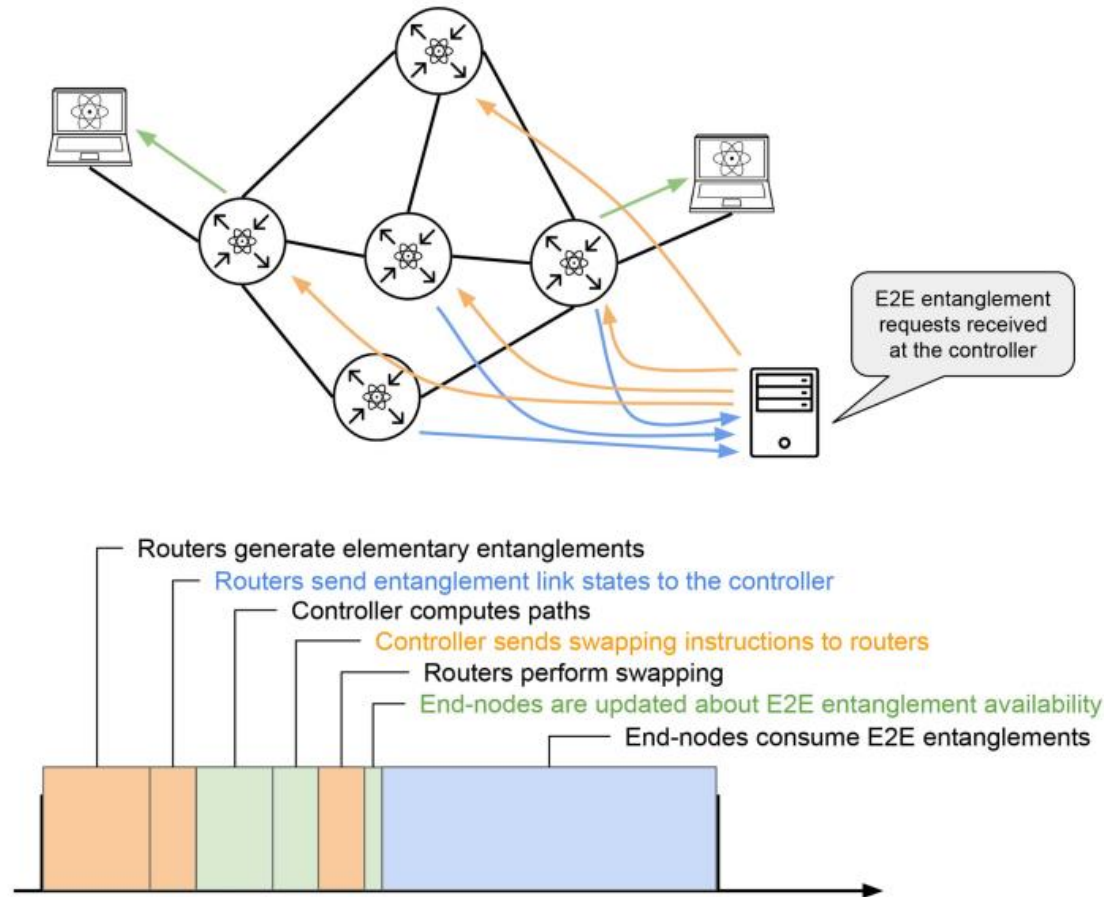
Distributed Proactive routing



Quantum Routing - Reactive

- In reactive routing (sometimes designated as advance or continuous entanglement generation), entanglements are constantly produced over each quantum channel across the network, and path computation is done on the instant logical topology formed by the created entanglements.
- This scheme frequently assumes a synchronous network where the quantum and routing operations evolve sequentially within discrete time slots of a fixed duration. Such a time-slotted system is used particularly with centralized path computation, although it can also be used in distributed routing.
- For centralized slotted reactive routing, at the beginning of a time slot, each pair of adjacent routers (i.e., sharing a quantum channel) attempts to generate entanglements. Entangled link states are communicated to the routing element (e.g., controller), which computes paths for the current E2E entanglement requests based on the logical topology formed by the created entanglements.

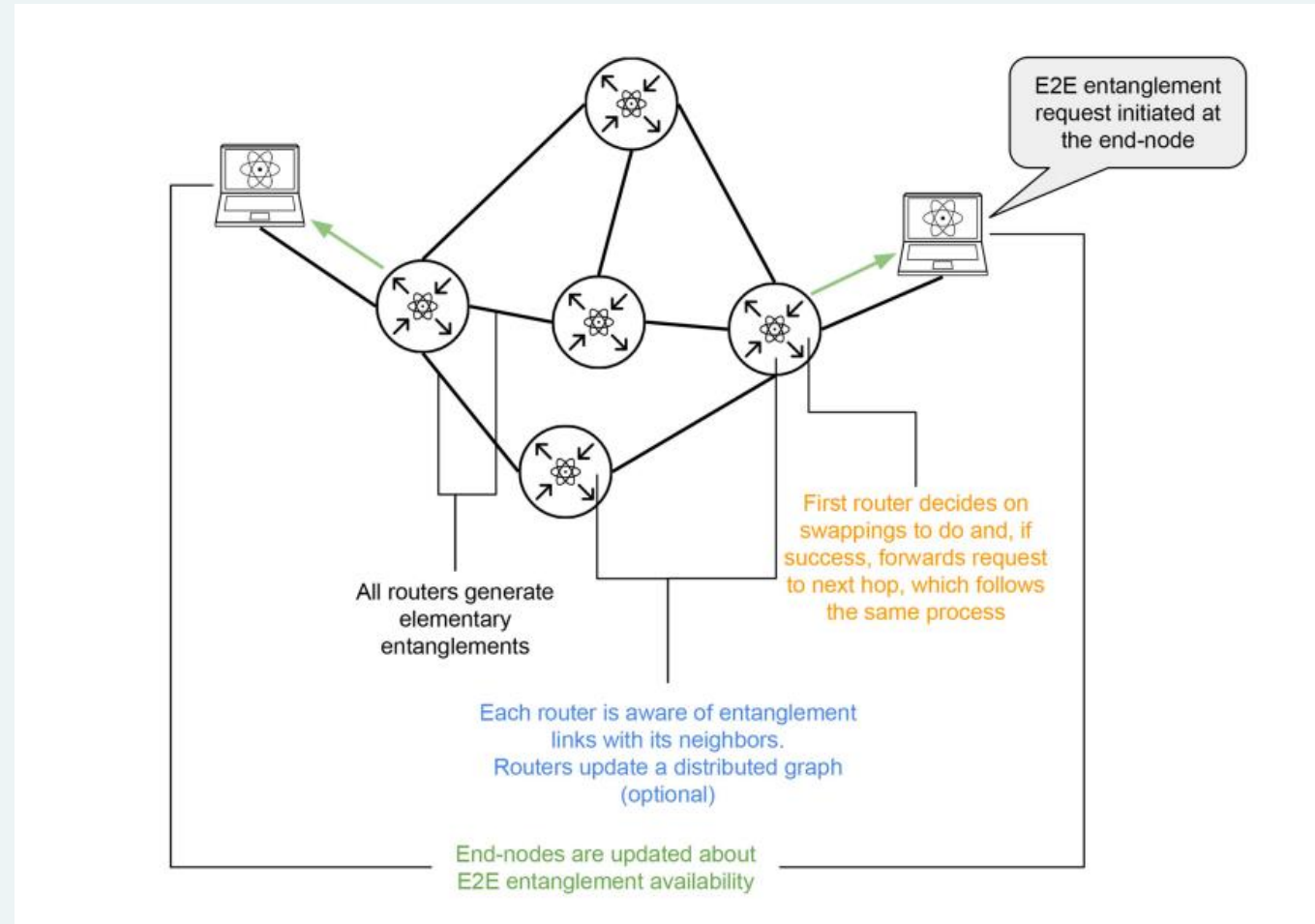
Centralised reactive routing



Quantum Routing – Reactive (Distributed)

- Reactive routing can also be designed in a distributed system, where routers rely only on the local knowledge of the logical topology to compute paths. The routing uses a path-finding mechanism in which all elementary entanglements are consumed in each attempt to establish an E2E entanglement.
- A more elaborated distributed path selection is where nodes collectively maintain the logical topology as a distributed graph.
- As new entanglements arrive and others expire, each node interacts with its neighbors to join the graph by selecting a root node based on link cost (e.g., fidelity). Once the node has joined a graph, it has a route toward the graph root, which is used to select entanglements to swap for each E2E entanglement request.

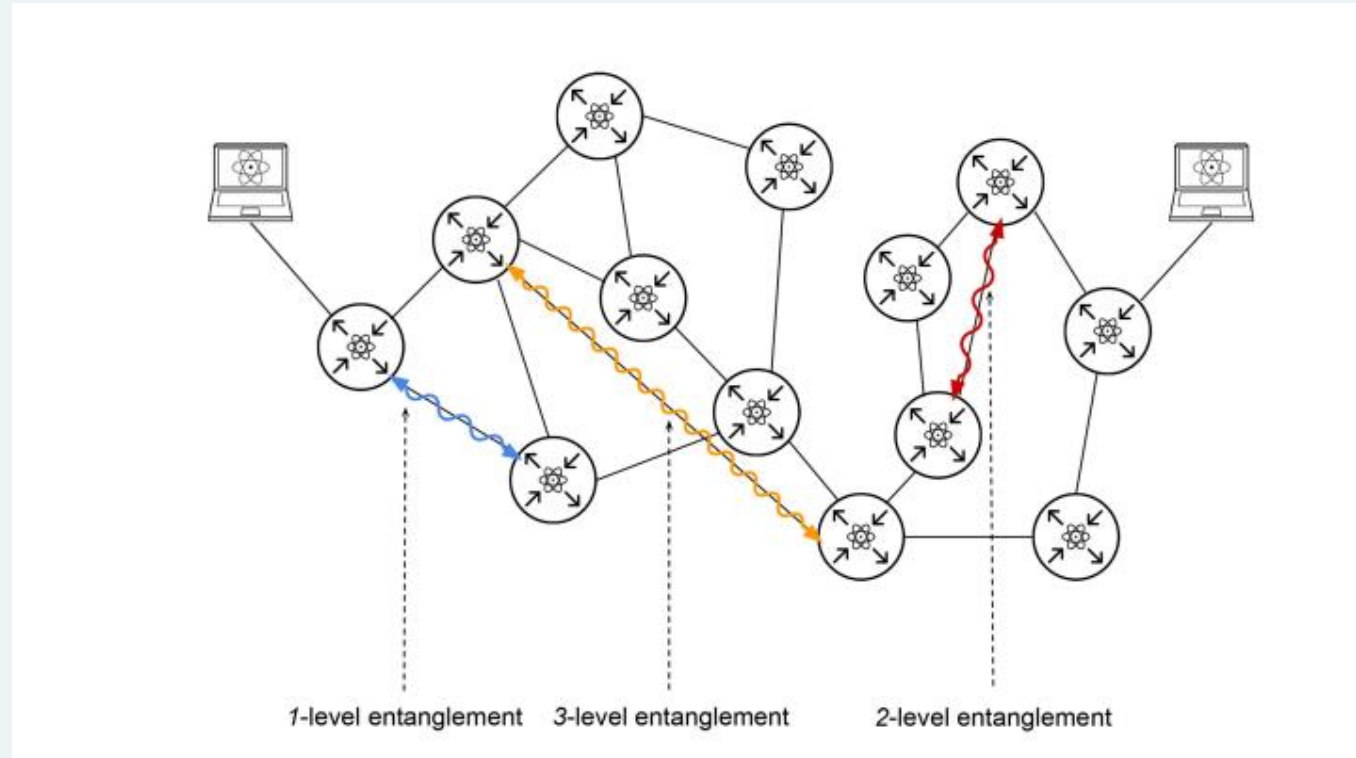
Distributed reactive routing



Quantum Routing – Hybrid

- Elements of proactive and reactive schemes can be combined to leverage their respective strengths while mitigating their weaknesses.
 - These hybrid approaches may use pre-shared entanglements or path selection concurrent to entanglement generation.
 - For instance, virtual routing pre-establishes virtual topologies using higher level entanglements to simplify path computation and reduce latency, while opportunistic routing dynamically selects paths at each node based on local information and ongoing entanglement attempts.
 - Virtual: Entanglements can be created between any pair of nonadjacent routers using swapping at intermediate nodes. This allows the network to form arbitrary virtual topologies with entanglements that extend beyond one physical link. Such entanglements are referred to as virtual links or i-level entanglements
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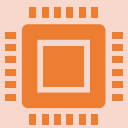
Hybrid Virtual routing – Various levels of entanglement links



Quiz 3: Quantum Routing



Contrast proactive and reactive quantum routing.



What advantages does hybrid quantum routing offer?

Quantum Network Simulators

- NetSquid: It is a modular and scalable quantum network simulator that models time-dependent processes such as qubit decoherence. It supports detailed hardware modeling and multiple quantum state representations, for efficient evaluation of quantum systems and network designs.
- SeQUeNCe: It is a full-stack discrete-event simulator designed to study quantum links, systems, and networks. It uses a notation (Bra-Ket) and density matrices, with a noise model that accounts for qubit decoherence, photon loss, and gate imperfections.
- QuISP: It is a simulation package for the quantum internet that implements the RuleSet protocol based on the OMNET++ simulator. The qubit model uses error tracking instead of full state representations.
- SimulaQron : It is a simulator for the application layer of quantum networks. Network layer software, such as routing protocols, can also be implemented and simulated.

Quantum Network Simulators

- QuNetSim: It is a high-level framework that simulates the network layer and above. It does not implement lower-level quantum device operations and thus is suitable for quick testing of quantum applications and routing protocols.
- ReQuSim: It is a lightweight Python-based quantum network simulator that simulates the entanglement distribution across quantum repeaters including loss and a wide range of imperfections such as memories with time-dependent noise.
- SimQN: It is a modular discrete-event-driven simulator specifically designed for evaluations of the network layer. SimQN aims to balance ease of configuration, scalability, and accuracy for simulating diverse quantum networking scenarios, including entanglement distribution, purification, and routing.

Summary

- Quantum Communication: Utilizes quantum bits (qubits) in superpositions to enable advanced computing and secure communication. Quantum states leverage entanglement, a unique quantum phenomenon, allowing secure quantum state transfers.
- No-Cloning and Decoherence Principles: Quantum states cannot be replicated exactly, making quantum communication secure yet challenging. Qubits are sensitive to environmental interactions (decoherence), affecting data integrity.
- Quantum Memory: Stores quantum information temporarily using photons and matter-based qubits, essential for long-distance quantum networking.
- Entanglement Purification and Quantum Error Correction (QEC): Techniques to enhance entanglement fidelity and correct errors, essential for reliable quantum communication.

Summary

- Quantum Repeaters and Swapping: Methods to extend quantum networks by distributing entanglements over long distances, overcoming optical losses and operational errors. It is important to note that the measurement process at the repeater destroys the two initial entanglements, thus requiring the generation of new elementary entanglements whenever swapping is used.
- Quantum Teleportation: Facilitates quantum state transmission without physical qubit transport, relying on entanglement and classical communication channels.
- Quantum Routing (Proactive, Reactive, and Hybrid): Strategies for efficiently managing quantum network resources to establish end-to-end entanglements.
- Quantum Network Simulators: Tools for modeling quantum network operations, such as NetSquid, SeQUeNCe, QuISP, SimulaQron, QuNetSim, ReQuSim, and SimQN.



THANK YOU